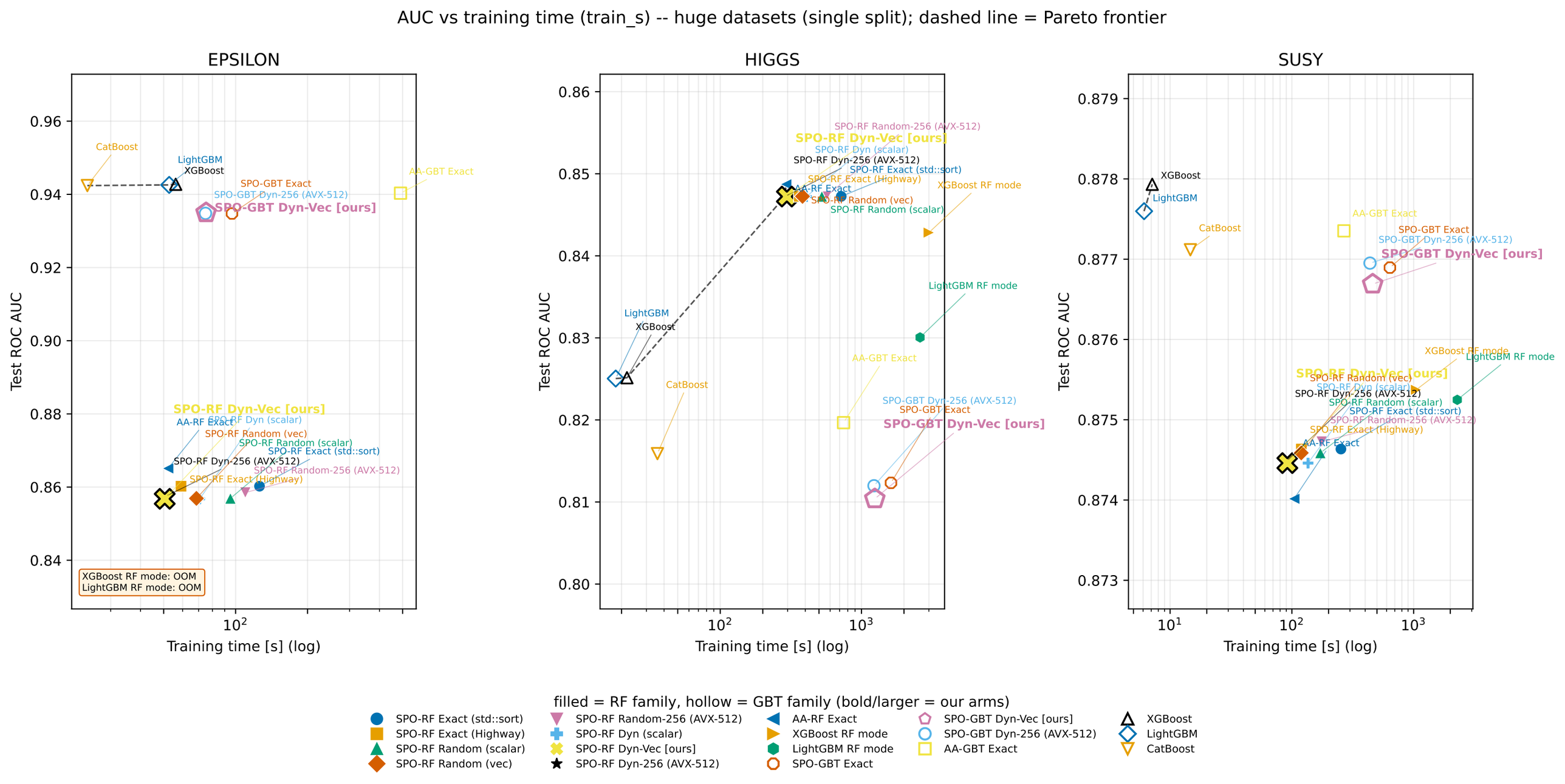
AUC vs training time (train_s) -- huge datasets (single split); dashed line = Pareto frontier
EPSILON
HIGGS
SUSY
Test ROC AUC
Training time [s] (log)
CatBoost
LightGBM
XGBoost
SPO-GBT Exact
SPO-GBT Dyn-256 (AVX-512)
SPO-GBT Dyn-Vec [ours]
AA-GBT Exact
SPO-RF Dyn-Vec [ours]
AA-RF Exact
SPO-RF Dyn (scalar)
SPO-RF Random (vec)
SPO-RF Random (scalar)
SPO-RF Exact (std::sort)
SPO-RF Dyn-256 (AVX-512)
SPO-RF Random-256 (AVX-512)
SPO-RF Exact (Highway)
XGBoost RF mode
LightGBM RF mode
XGBoost RF mode: OOM
LightGBM RF mode: OOM
filled = RF family, hollow = GBT family (bold/larger = our arms)
SPO-RF Exact (std::sort)
SPO-RF Exact (Highway)
SPO-RF Random (scalar)
SPO-RF Random (vec)
SPO-RF Random-256 (AVX-512)
SPO-RF Dyn (scalar)
SPO-RF Dyn-Vec [ours]
SPO-RF Dyn-256 (AVX-512)
AA-RF Exact
XGBoost RF mode
LightGBM RF mode
SPO-GBT Exact
SPO-GBT Dyn-Vec [ours]
SPO-GBT Dyn-256 (AVX-512)
AA-GBT Exact
XGBoost
LightGBM
CatBoost